# JADY PAMELLA

AI Engineer Focused on Digital Twins and AI Safety | PhD Fellow, Aarhus University | EA Sweden Fellow 2026 | BlueDot Impact

AI Engineer focused on AI safety who builds trustworthy production AI. 12 years shipping AI inside regulated environments (a public-sector bank, federal research labs, and climate tech), from LLM platforms and RAG pipelines to computer vision and MLOps at safety-critical latency budgets. PhD at Aarhus University on digital twins for district cooling (SWiM, funded by the Grundfos Foundation) adds depth in AI safety, cybersecurity, and critical-infrastructure resilience. Peer-reviewed in Elsevier Computers and Security.

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## Fellowships

**EA Sweden Fellowship**
Effective Altruism Sweden · 2026 cohort

**AGI Strategy and Technical AI Safety**
BlueDot Impact · 2026 · Future of AI completed 2025

**Network for Global Professionals**
Swedish Institute · 2024 to 2026

## Education

**Stockholm University, DSV**
MSc Computer and Systems Sciences (AI and IoT). Aug 2024 to May 2026. Stockholm, Sweden. Swedish Institute Scholar (SISGP).

**University of Brasília (UnB)**
Professional Master's in Electrical Engineering, Cybersecurity Area (PPEE). Mar 2024 to May 2026. Brasília, Brazil.

**UPIS Faculdades Integradas**
BSc Information Systems. 2010 to 2013. Brasília, Brazil.

## Volunteer Work

**Student Ambassador**
Drivhuset Stockholm · 2025 to 2026 · Stockholm, Sweden

**Volunteer Frontend Developer**
Stiftelsen Effekt / Ge Effektivt · geeffektivt.se · 2026

**Web Developer**
Alla Ska Få Mat · 2024 to Present · Stockholm, Sweden

## Publications

Alves, R. S., **Barbacena da Silva, J. P.**, Ribeiro Jr., L. A., Rabelo Nunes, R. (2025). *Enhancing Cybersecurity in the Judiciary: Integrating Additional Controls into the CIS Framework.* Computers and Security (Elsevier), IF 5.6. DOI 10.1016/j.cose.2025.104584.

**Barbacena da Silva, J. P.**, et al. (2024). *Cybersecurity Protection in the Brazilian Judiciary.* NAVUS, Journal of Technology and Management. DOI 10.22279/navus.v14.2032.

**Barbacena da Silva, J. P.**, et al. (2025). *Critical Infrastructure Under Threat: Business Risks for Cybersecurity in Metropolitan Transport Systems.* ENANGRAD 2025 (Brazilian National Meeting of Administration Programs).

**Barbacena da Silva, J. P.**, et al. (2026). *Information Security Controls Baseline for Mitigating Business Risks in the Sanitation Sector.* ENANGRAD 2026. Paper award.

## Languages

- Portuguese (Native)
- English (C1, TOEFL 101/120)
- Spanish (B1)
- Swedish (Beginner)
- Danish (Beginner)

## Experience

**Aarhus University, PhD Fellow, Digital Twins for District Cooling (SWiM)** — Aug 2026 to Present (Aarhus, Denmark)

- Verifying that **district cooling installations are commissioned as they were designed**, combining Building Information Modelling, AI and digital twins. Part of SWiM, a five-year project on sustainable water-based cooling for megacities funded by the Grundfos Foundation, supervised by Peter Gorm Larsen and Carl Schultz.

**Deep Forestry AB, AI Research Engineer** — Dec 2025 to Jun 2026 (Uppsala, Sweden)

*MSc thesis on safe landing-zone detection for autonomous drones in below-canopy forests.*

- Built a hybrid geometric and neural pipeline (RANSAC, PointNet, PointNet++, DGCNN) reaching **F1 0.926 at 53 ms per frame on NVIDIA Jetson Orin NX**, a **29× latency reduction (5.8s to 200ms)** over the deep-learning baseline.
- Combined KD-Tree indexing and voxel downsampling with the neural stack to hit the edge latency budget. Counterfactual gain validated with session-level cross-validation on real LiDAR flight data.

**SoiQet, Founder and CEO** — Aug 2025 to Present (Aarhus, Denmark)

*AI-powered social media platform for creators, brands, and teams (Plug Into Social Intelligence), from prototype to production.*

- Applied AI-engineering practices across **6 LLM providers** (OpenAI, Anthropic, and others), including model evaluation, so model-swap and build-vs-buy decisions are measurable (output stability and structured-output adherence) instead of guesswork.
- Shipped production in-house RAG with structured outputs, guardrails, OAuth across 6 platforms, Stripe billing, and Supabase row-level security. Chose RLS over Firebase because B2B multi-tenant access boundaries had to be auditable.
- Assembled and led the team, set the roadmap, and own delivery from build to launch.

**Independent, Technical Advisor (AI-powered consumer product)** — 2026 (Remote)

- Delivered a **60-page technical audit** of an AI-powered consumer product MVP. Identified **20 of 21 Supabase tables without RLS** (verified exposure with public API key), **29 missing AI and security guardrails**, and AI cost runaway risk. Delivered product strategy reframe plus 8-week prioritised roadmap.

**LabRisk, University of Brasília, Cybersecurity Research Engineer** — Aug 2024 to Present (Brasília, Brazil)

- Research on risk management, AI for critical-infrastructure cybersecurity, and security governance frameworks. Output includes the 2025 Elsevier *Computers and Security* paper extending the CIS framework for the Brazilian judiciary, and the 2025 ENANGRAD conference paper on cybersecurity business risks in metropolitan transport systems.

**Bank of Brasília (BRB), IT Analyst and DevOps Engineer** — Dec 2014 to Present (Brasília, Brazil)

- Delivered a production NLP chatbot (Python, TensorFlow) that **cut response time 40% for 100K+ users**, then led the build-vs-buy pivot that routes **40% of BRB customer service through the resulting WhatsApp channel**. Built GDPR-aligned analytics for the instant payments system on PySpark.
- Rebuilt the release pipeline on Jenkins, GitLab, Docker, Kubernetes, OpenShift. **Deployment time cut 60%**. Held the rollout for 2 months to close the compliance gaps first, in a regulated banking environment.

**Bank of Brasília (BRB), IT Manager and Engineering Manager** — May 2016 to Jul 2024 (Brasília, Brazil)

- Owned a **$1M+ annual IT budget** and a **team of 10+ engineers**. Cut operating cost **15%**, saved **$150K+ in vendor renegotiation**, lifted audit accuracy **40%** in a regulated banking environment.

## Awards and Recognition

- **1st place, EpiMinds Google AI Hackathon (2026).** Shipped a Multi-Agent Crisis Response System (MACS) in a single day.
- **1st place, Kong and Kolomolo AI-Assisted Coding Hackathons (2025).**
- **Paper award, 37th ENANGRAD (2026).** Security controls baseline for the sanitation sector.
- **Reviewer, IRC-KDU 2026.** International Research Conference, General Sir John Kotelawala Defence University, Sri Lanka.
- **Judge, Apart Research Global South AI Safety Hackathon (2026).** Technical AI Safety, Governance and Security tracks.
- **Speaker, Nordic Tech Week "New Tech, Old Rules" panel (2026).**